

Professional Practices

Lecture 7b

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CASE STUDY

- A software company is developing a healthcare application. During testing, the testing team discovers that the system occasionally produces erroneous reports. Fixing the issue would delay the project by five weeks. The project manager, under pressure from the client and higher management, instructs the team to **ignore the issue temporarily** and deliver the product on time. He argues that:
 - “We can fix it later in updates, but missing the deadline will damage our reputation.”
 - **Is it ethically correct to release software with known defects?**
 - **What are the risks for patients?**
 - **Should the team follow the manager’s instructions or report the issue?**

Is it ethically correct to release software with known defects?

- No, it is **not ethically correct**. Reasons:
- **Duty of care:** Healthcare software directly affects patient health. Releasing software with known defects risks harming patients.
- **Professional integrity:** Software engineers and QA professionals have a responsibility to ensure their products are reliable, especially in critical domains like healthcare.
- **Deceptive practices:** Delivering a defective product knowingly is misleading to the client and end users.

What are the risks for patients?

- **Misdiagnosis or delayed treatment:** Incorrect reports may lead to wrong treatments or missed diagnoses.
- **Health deterioration:** Patients may receive inappropriate medications or miss necessary interventions.
- **Loss of trust:** Patients and healthcare providers may lose confidence in the software and the healthcare system.
- **Legal consequences:** Errors may lead to malpractice claims against healthcare providers and liability for the software company.

Should the team follow the manager's instructions or report the issue?

- **The team should prioritize patient safety and professional ethics over deadlines.** Steps they should take:
- **Document the issue:** Clearly note the defect and its potential impact.
- **Escalate responsibly:** Report the issue to higher management, the client, or a compliance officer, emphasizing patient safety.
- **Refuse unsafe release:** Releasing the software with known critical defects would violate ethical and possibly legal standards.
- **Suggest alternatives:** Propose a plan to release a limited or non-critical version, or delay the release until the defect is fixed.
- **Key point:** In healthcare software, **patient safety always takes precedence over deadlines or reputation.**

Contract and Delivery for a Software

- **Software Product**

The Contractor shall deliver a fully functional Learning Management System (LMS) that meets the specifications outlined in Appendix A (System Requirements Specification). The LMS shall include, but is not limited to, the following modules and functionalities:

- **User Management:** Registration, authentication, role assignment (student, faculty, staff, admin), and profile management.
- **Course Management:** Creation, modification, and deletion of courses; syllabus upload; enrollment management.
- **Content Management:** Upload, download, and organization of course materials including videos, PDFs, and other digital content.
- **Assessment & Grading:** Online quizzes, assignments submission, automated grading (where applicable), and gradebook management.
- **Communication Tools:** Announcements, discussion forums, messaging, and notifications.
- **Reporting & Analytics:** Student performance reports, course engagement statistics, and administrative dashboards.
- **Integration Capabilities:** Support for integration with existing university systems (e.g., student information systems, email services).

- **Documentation**

The Contractor shall deliver complete documentation, including:

- **User Manuals:** Detailed instructions for end users (students, faculty, staff).
- **Administrator Guides:** Instructions for system administrators to manage, configure, and maintain the LMS.
- **Technical Documentation:** System architecture, API references, database schema, and source code documentation sufficient for future maintenance or enhancements.

- **Testing & Quality Assurance**

The Contractor shall provide test plans, test cases, and test results demonstrating that the LMS meets the agreed functional and non-functional requirements. This includes:

- Unit and integration testing reports.
- User acceptance testing (UAT) results.
- Bug tracking and resolution logs.

- **Deployment & Training**

The Contractor shall deliver:

- A fully deployed LMS on the university's chosen hosting environment or cloud platform.
- Training sessions for faculty, staff, and administrative personnel to ensure effective use of the system.
- Support for an agreed period post-deployment for bug fixes and minor enhancements.

- **Source Code & Ownership**

Upon completion and acceptance, the Contractor shall deliver all source code, libraries, and associated assets developed for the LMS. Ownership and intellectual property rights shall be as defined in Section X (Intellectual Property Rights).

Actions to Establish and Promote an Ethical Culture

- **Implement Clear Policies and Codes of Conduct**
 - Develop and communicate a formal **Code of Ethics** covering fair treatment, workload management, transparency, and professional behavior.
 - Ensure policies explicitly discourage overwork without proper compensation.
- **Promote Work-Life Balance**
 - Enforce reasonable working hours and discourage mandatory overtime.
 - Introduce flexible schedules or remote work options where possible.
- **Transparent Communication and Feedback Mechanisms**
 - Hold regular meetings to discuss workload, project progress, and employee concerns.
 - Establish **anonymous feedback channels** or an ethics hotline for reporting unfair practices.
- **Recognition and Reward Systems**
 - Implement **fair compensation and performance-based incentives**.
 - Recognize high performers not only for output but also for collaboration, ethics, and contribution to team growth.
- **Leadership by Example**
 - Demonstrate ethical behavior at the management level—e.g., respecting deadlines without enforcing excessive overtime.
 - Encourage managers to model empathy, fairness, and professional integrity.
- **Training and Awareness**
 - Provide **ethics training** and workshops to employees on professional responsibility, workplace fairness, and organizational values.

Factors to Consider for Long-Term Career Planning Strategy

- **Employee Skills and Aspirations**
 - Understand each employee's **career goals, strengths, and areas for development** through performance appraisals and one-on-one sessions.
- **Growth Opportunities**
 - Identify **clear career paths** within the company for technical, managerial, or specialized roles.
 - Offer **mentorship and coaching programs** to prepare employees for future roles.
- **Training and Development Programs**
 - Provide access to **skill development courses, certifications, workshops**, and cross-functional projects.
- **Succession Planning**
 - Prepare employees to take on higher responsibilities in critical positions, ensuring continuity in leadership and technical expertise.
- **Work-Life Balance and Retention**
 - Ensure career progression plans include **balanced workloads** and **recognition of contributions**, reducing burnout risk.
- **Performance Metrics and Feedback**
 - Use **transparent performance evaluation systems** aligned with promotions, raises, and career opportunities.
- **Alignment with Organizational Goals**
 - Ensure career paths and growth plans support **long-term organizational objectives**, creating mutual benefit for employees and the company.

Types of Intellectual Property

- **Trade Secrets**

- The unique algorithm developed by DataNova is a **trade secret** because it provides a competitive advantage and is meant to remain confidential.
- Trade secrets are protected as long as the company takes reasonable steps to maintain their secrecy.

- **Copyright**

- The **user interface design, code, and documentation** are protected under copyright law.
- Copyright safeguards original expressions (code, graphics, UI layouts) but not underlying ideas or algorithms unless they are published or patented.

- **Potential Patents**

- If the algorithm is novel, non-obvious, and useful, it could be **patentable**. A patent would give the company exclusive rights to use or license the algorithm for a limited period.

- **Possibly Trade Dress**

- If the product's interface is distinctive and identifies the source of the software, **trade dress protection** could apply to the UI design.

Strategies to Protect Trade Secrets

- **Non-Disclosure Agreements (NDAs)**

- Require all employees, contractors, and collaborators to sign NDAs covering proprietary information, algorithms, and design details.

- **Access Control and Confidentiality Policies**

- Limit access to sensitive algorithms or source code to only essential personnel.
- Use secure systems and authentication to prevent unauthorized sharing.

- **Employee Agreements and Exit Protocols**

- Include clauses in employment contracts that restrict former employees from sharing or using trade secrets after leaving the company.

- **Internal Documentation of Measures**

- Keep records showing that reasonable steps were taken to maintain confidentiality. This is crucial for legal protection if a dispute arises.

Preventive Measures

- **File a Patent or Provisional Patent**

- Protect the unique algorithm legally to prevent competitors from using it, even if an employee leaves.
- Patents give enforceable rights for 20 years (in most jurisdictions) and can also attract investors.

- **Implement Strong Legal Agreements and Policies**

- NDAs, non-compete clauses (where legally enforceable), and explicit IP ownership agreements could prevent ex-employees from replicating the product.